Alur Ankitha

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	7.52
Sri Chaitanya Narayana Junior College	Class XII (Telangana State Board of Intermediate Education)	2019	95.20%
Shine High School	Class X (Telangana State Board of Secondary Education)	2017	9.80%

PROJECTS

Quantum Chaos Classification via ML

- **Built an exact-diagonalization and ML pipeline** to classify integrable vs chaotic regimes of the mixed-field Ising model using spectral statistics, including level-spacing distributions; achieved **90% test accuracy (AUC 0.94)** with Random Forest.
- **Resolved energy spectra by symmetry sectors** to reduce spurious level correlations, and identified low-spacing statistics as dominant physical features through Random Forest feature-importance analysis.
- **Tested cross-system-size generalization** by training on L=8,10 spin chains and evaluating on unseen L=12 systems, probing whether learned spectral signatures of quantum chaos persist beyond the training system size.

Quantum Simulation -- Reproducing Zalka (1998)

- Independently derived and implemented **first- and second-order split-operator evolution**; verified **Trotter error scaling** (1.000, 2.023) and norm conservation to 10^{-12}
- Extracted **harmonic-oscillator energy spectrum** via **autocorrelation Fourier analysis** and validated recovered eigenvalues against analytical results
- Extended the solver to **double-well potentials** to study **tunnelling dynamics** and low-lying energy splitting
- Implemented a **QFT-based gate-level realization** for small registers and benchmarked quantum-circuit evolution against the classical FFT solver

Open-System Exciton Transfer Dynamics

- Modeling excitonic energy-transfer dynamics in photosynthetic systems using **open quantum-system frameworks**, with emphasis on system-environment interactions and decoherence.
- Developing computational models to study the influence of **environmental coupling, energy landscapes, and inter-site interactions** on exciton populations, coherence, and transfer efficiency.
- Exploring **reinforcement-learning-based control strategies** to optimize exciton transfer under environmental fluctuations and identify robust dynamical pathways.

Gravitational Wave & Neutron Star Data Analysis

- **Reproduced the GW150914 LIGO signal-processing pipeline** from open H1/L1 strain data using Welch PSD estimation, frequency-domain whitening, Butterworth bandpass filtering, Q-transform analysis and matched filtering; recovered **network SNR 30.7**.
- **Quantified pipeline sensitivity to filter cutoffs, template chirp-mass mismatch and whitening-window length**, demonstrating how preprocessing and waveform mismatch affect gravitational-wave SNR and signal recovery.
- **Compared Gaussian-mixture models for 15 Galactic double-neutron-star masses using AICc, BIC, Bayesian marginal likelihood and leave-one-out cross-validation**, showing that the available sample provides insufficient evidence for a two-component population model.

POSITIONS OF RESPONSIBILITY

Placement Coordinator | OCS IITH

- **Led corporate outreach to multiple companies** for internship and placement opportunities, pitching IIT

Hyderabad's talent pool and recruitment ecosystem
- **Managed company coordination and recruitment communication**, engaging with corporate representatives to understand hiring requirements and facilitate candidate opportunities
- **Screened and verified CVs/profiles for 7 assigned students**, assessing alignment with company-specific eligibility and role requirements

Sahaay Team Coordinator & SMC Mentor

- Coordinated Sahaay (Social Innovation Club, IITM), coaching student teams on **early-stage startup development**; organized CFI Open House with **2000+ attendees** and produced an accessible explainer video of technical work
- Mentored **5 first-year students** (Saathi Mentorship Council) through one-on-one sessions, supporting academic and professional development during the pandemic

UNDERGRADUATE PROJECTS

FeelSafe Women's Safety Device

- Led **PCB design** and hardware integration as **Electronics Lead**; **compressed multi-feature electronics** (SOS, live location tracking) into a wristwatch-form-factor device using Eagle CAD and Fusion 360
- Conducted a **150-response survey and 15 user interviews** to validate design requirements; applied **Disciplined Entrepreneurship** framework to guide project scope and market positioning

SKILLS

- **Programming:** Python (NumPy, SciPy, Matplotlib, scikit-learn); QuTiP; Qiskit
- **Physics methods:** Exact diagonalisation, Trotter-Suzuki decomposition, Lindblad master equations, random matrix theory, matched filtering, Welch PSD estimation
- **Machine learning:** SVM (RBF/polynomial/linear), Random Forest, Logistic Regression, ROC/AUC, model selection (AICc, BIC, LOO-CV)

RELEVANT COURSES

ML for Physical Sciences, Quantum Computing, Quantum Sensing, Quantum Devices, Solid State Devices, Data Science Analysis

EXTRACURRICULAR

Hult Prize 2020 -- Farm-to-Home Food Supply Platform

- **Developed and pitched a farm-to-home startup concept** connecting small and marginal farmers with consumers, focusing on technology-access barriers and agricultural supply-chain inefficiencies
- Defined the **product concept, user flows, and proposed feature set**, including farmer enrollment, consumer access, and crop-specific weather and soil-suitability information